

TOPICAL REVIEW

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TOPICAL REVIEW

Ultrasound contrast microbubbles in imaging and therapy: physical principles and engineering

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Abstract

Microbubble contrast agents and the associated imaging systems have developed over the past 25 years, originating with manually-agitated fluids introduced for intra-coronary injection. Over this period, stabilizing shells and low diffusivity gas materials have been incorporated in microbubbles, extending stability *in vitro* and *in vivo*. Simultaneously, the interaction of these small gas bubbles with ultrasonic waves has been extensively studied, resulting in models for oscillation and increasingly sophisticated imaging strategies. Early studies recognized that echoes from microbubbles contained frequencies that are multiples of the microbubble resonance frequency. Although individual microbubble contrast agents cannot be resolved—given that their diameter is on the order of microns—nonlinear echoes from these agents are used to map regions of perfused tissue and to estimate the local microvascular flow rate. Such strategies overcome a fundamental limitation of previous ultrasound blood flow strategies; the previous Doppler-based strategies are insensitive to capillary flow. Further, the insonation of resonant bubbles results in interesting physical phenomena that have been widely studied for use in drug and gene delivery. Ultrasound pressure can enhance gas diffusion, rapidly fragment the agent into a set of smaller bubbles or displace the microbubble to a blood vessel wall. Insonation of a microbubble can also produce liquid jets and local shear stress that alter biological membranes and facilitate transport. In this review, we focus on the physical aspects of these agents, exploring microbubble imaging modes, models for microbubble oscillation and the interaction of the microbubble with the endothelium.

(Some figures in this article are in colour only in the electronic version)

¹ These authors contributed equally to this review.

1. Introduction

The engineering of ultrasound contrast agents, co-optimized with systems for imaging their local distribution and enhancing their use in local drug delivery, is a rich, multidisciplinary undertaking. Ultrasound contrast agents are small gas bubbles encapsulated by a stabilizing shell, with a typical diameter on the order of microns (table 1). These microbubbles are injected intravenously and remain within the blood pool, with early agents shown to circulate in a manner similar to red blood cells (Jayaweera *et al* 1994). Ultrasound pulses are typically applied with a frequency near the resonance frequency of the gas bubble and the bubbles increase and decrease in diameter, producing strong echoes from regions of perfused tissue (de Jong *et al* 1994b, Dayton *et al* 1999b). Trains of ultrasound pulses with varied frequency, phase and amplitude are designed to separate bubble and tissue echoes under well-characterized conditions and we review these pulse trains here. Optimized pulse trains for quantitative imaging of blood vessel density and perfusion and enhanced drug delivery continue to evolve.

Several generations of the microbubble agents have also been developed, where early agents contained an air core and were stabilized by a coating of albumin, starting with Albunex®. Agents with a fluorinated gas core were then developed, including Optison™ (GE Healthcare Systems) with a protein shell and perfluoropropane gas core and Definity® (Bristol-Myers Squibb) with a phospholipid shell and perfluoropropane core (Definity-Prescribing-Information; Optison-Prescribing-Information). Microbubbles are typically manufactured by mechanical agitation, although microfluidic methods to engineer precise size distributions are in development (Talu *et al* 2006). An excellent history of microbubble agents was recently provided (Kaul 2008), the biophysics of the agents and their shells was detailed in Ferrara *et al* (2007) and the chemistries of the agents were described in greater detail in Klibanov (2005). Within this review, we focus on the engineering and physical principles behind the imaging systems that employ microbubbles, models that have been developed to describe microbubble response to an ultrasound pulse and the impact of microbubble oscillation on small blood vessels.

2. Experimental studies of oscillation, echoes and spectra

In response to an ultrasonic pressure wave, microbubbles expand and contract with each cycle of pressure. Optical imaging methods to visualize the increasing and decreasing diameter have been developed (Klibanov *et al* 1998, Dayton *et al* 1999b, Chomas *et al* 2000, Postema *et al* 2005, Garbin *et al* 2007), where two-dimensional images of the microbubble can be acquired with shutter speeds on the order of nanoseconds. Alternatively, images of a single line through the center of a bubble are recorded, where these recordings, known as ‘streak’ images, can be updated with nanoseconds or picoseconds between recordings (Chomas *et al* 2000). For a typical microbubble with a diameter of $\sim 2 \mu\text{m}$, the streak image acquired during insonation with a 2.25 MHz center frequency pulse at a low peak rarefactional pressure (PRP) ($\sim 100 \text{ kPa}$) demonstrates the increasing and decreasing bubble diameter (figure 1(a)). For a low transmitted PRP, the microbubble oscillation corresponds to the rarefaction and compression of the ultrasound wave, with a frequency close to the incident ultrasound frequency (figures 1(a), (c)). Within figure 1(c), predictions for the expansion ratio (defined as the instantaneous radius normalized by the initial radius) are illustrated for two transmitted PRP amplitudes of 30 and 60 kPa. For these very low values, oscillation is nearly sinusoidal with the expansion scaling nearly linearly (two-fold) between these values.

With a higher PRP, the magnitude of the expansion and the speed of microbubble collapse each increase substantially, and the oscillation of the microbubble has a nonlinear relationship

Table 1. Physical characteristics and dosage recommendations for ultrasound contrast agents used in imaging and drug delivery studies.

Formulation	Shell	Gas	Concentration (mL ⁻¹)	Mean diameter (μm)	Recommended dose ^a	Reference
Optison	Albumin	C ₃ F ₈	5.0–8.0 × 10 ⁸	3.0–4.5	0.5 mL	(Optison- Prescribing- Information)
Definity	Lipid	C ₃ F ₈	1.2 × 10 ¹⁰	1.1–3.3	10 μL kg ⁻¹	(Definity- Prescribing- Information)
PESDA	Dextrose albumin	C ₄ F ₁₀	6.5 × 10 ⁸	2.5–4.9	2.5–10 (μL kg ⁻¹)	(Porter <i>et al</i> 1996)

^a Bolus intravenous injection into the peripheral vein.

to the driving pressure (figures 1(b), (d)) (Morgan *et al* 2000). The nonlinear nature of microbubble oscillation is again illustrated by the predicted expansion ratios, where doubling of the transmitted PRP from 150 to 300 kPa increases the peak expansion ratio by more than 2.5-fold (figure 1(d)). Also, for these higher values of transmitted PRP, microbubble expansion is followed by a comparatively faster collapse. As a result, the frequency spectra of echoes produced by the microbubble contain multiples and in some cases a sub-multiple of the transmitted frequency (Schrope and Newhouse 1993, de Jong *et al* 1994b, Chomas *et al* 2002). These harmonic echo components are used to detect the presence and concentration of microbubbles throughout the body, as described in section 4. The rate of microbubble expansion and collapse depends upon the sound field as well as the diameter of the microbubble (as compared with the transmitted pulse) and nearby boundaries.

Raising the PRP to 300 kPa (with all other parameters held constant) also results in the destruction of the microbubble during the pulse, where asymmetries and fragments are visualized (figure 1(b)). The mechanisms of microbubble destruction are described in detail in Chomas *et al* (2001a, 2001b) including diffusion of the gas from the microbubble core in the absence of ultrasound, enhanced diffusion of the gas from the microbubble core in the presence of ultrasound and the fragmentation of the microbubble during oscillation (figure 2(a)). Recordings of the echoes that result from oscillation and microbubble destruction demonstrate an increased center frequency as the diameter decreases and decreased echo amplitude with microbubble fragmentation (figure 2(b)).

Expansion and fragmentation of lipid-shelled microbubbles are determined primarily by the resting microbubble diameter, frequency, phase and amplitude of the ultrasound pulse (figure 3). Increasing PRP, decreasing frequency and decreasing resting diameter increase the normalized expansion and likelihood of microbubble fragmentation. For a lipid-shelled microbubble, the lipid molecules self-assemble as a coating on the gas–liquid interface (Kim *et al* 2000, Lee *et al* 2001, Borden and Longo 2002, Borden *et al* 2005a, 2005b, 2006a). The thin, lipid shell buckles into a bilayer during compression and leaves unprotected regions of the interface during expansion. Alternatively, polymer and albumin shells can rupture during oscillation with discrete defects visualized (Bloch *et al* 2004, 2005, Patel *et al* 2004).

Although harmonic echoes are often associated with microbubbles, tissue echoes also contain harmonic multiples of the initial transmitted center frequency, resulting from the nonlinear propagation of the transmitted ultrasound pressure wave. The speed of sound is higher for the higher pressure components of an ultrasound wave traveling within tissue and therefore the pulse distorts during propagation. Tissue echoes from the distorted pressure wave also contain harmonic frequencies (Schoelgens 1998).

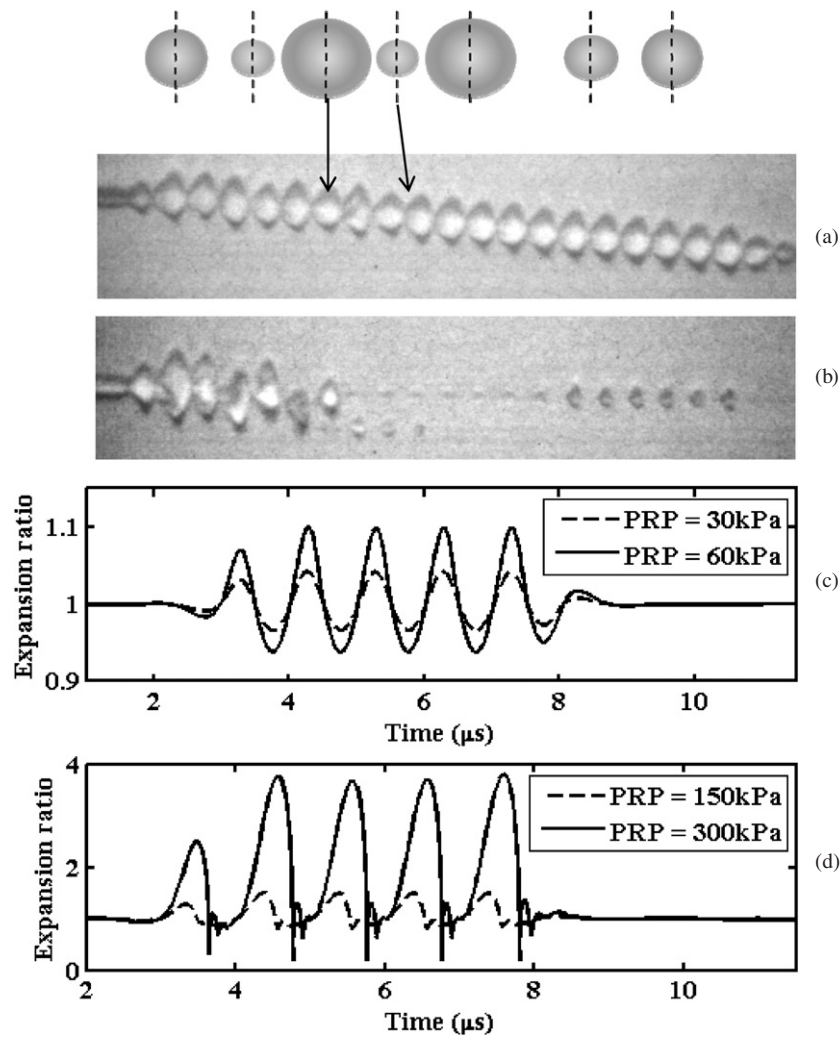


Figure 1. Observed and predicted microbubble oscillation. (a) and (b) Experimental streak images of $1.4\ \mu\text{m}$ lipid-shelled microbubbles insonified with a 2.25 MHz pulse with a PRP of ~ 100 (a) or 300 kPa (b). The streak image is an optical image acquired with an ~ 15 ns shutter showing a single line through the center of the microbubble (shown as dashed line in cartoon) as it expands and contracts during the ultrasound pulse. Momentum is transferred to the microbubble from the ultrasound pulse resulting in a net displacement of the microbubble. At the higher ultrasound pressure, the microbubble fragments and the resulting small microbubbles can be visualized. (c) and (d) Predicted microbubble expansion ratio versus time for a bubble with an initial size of $2\ \mu\text{m}$ during insonation with 1 MHz five-cycle ultrasonic pulses using the equation and parameters as in Zhao *et al* (2004). (c) For a low transmitted PRP, microbubble oscillation is harmonic at a frequency close to the incident ultrasound frequency and the bubble expansion ratio is nearly proportional to the transmitted PRP. (d) For a high transmitted PRP, the oscillation nonlinearity increases and the expansion ratio increases more rapidly than the transmitted PRP.

With a bubble oscillating near a boundary, asymmetric collapse occurs and fluid jets can impinge on and damage a gel surface (figure 4(a)) (Kodama and Tomita 2000). Although the cavitating bubbles in figure 4(a) are large (diameter of 2 mm) compared to microbubbles, jet-like formations have been observed with microbubbles driven at low ultrasonic transmission

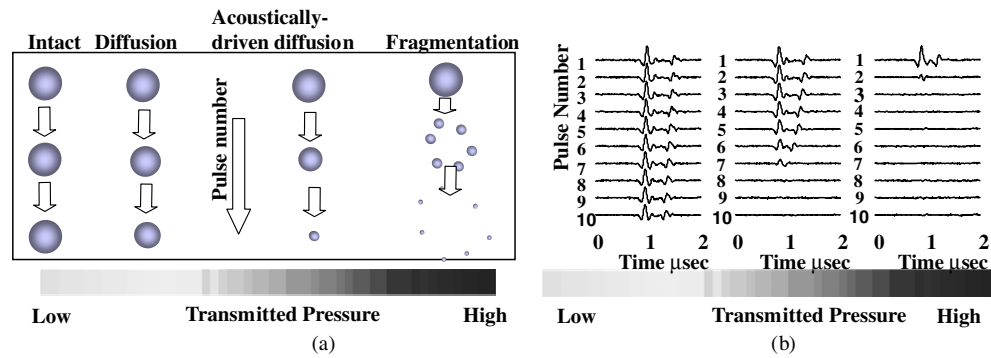


Figure 2. (a) Cartoon demonstrating the mechanisms of ultrasound contrast agent destruction, including diffusion, acoustically-driven diffusion and fragmentation. (b) Recordings of the echoes from single microbubbles during insonation with a train of 5.0 MHz pulses with a pulse duration of 1.5 cycles, pulse repetition frequency of 1 kHz and varied PRP. With a low PRP, similar echoes are obtained for 10 pulses; with a high PRP, the microbubble fragments after a single pulse.

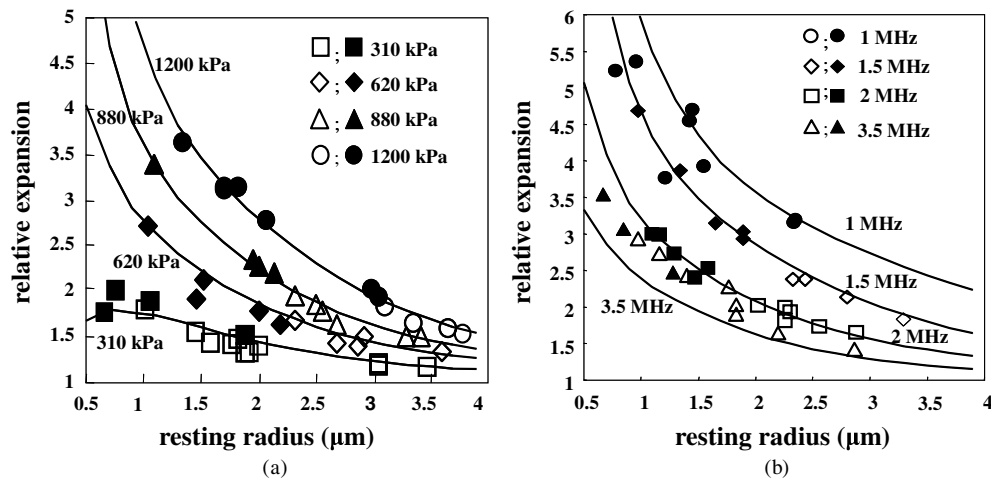


Figure 3. (a) Relative expansion of microbubbles during insonation with a two-cycle sinusoid at 2.25 MHz with a PRP ranging from 310 kPa to 1200 kPa. (b) Relative expansion of microbubbles during insonation with a two-cycle sinusoid with a PRP of 800 kPa

frequencies (~ 1 MHz) (Prentice *et al* 2005). Fluid jets produced by oscillating microbubbles are predicted to be spatially smaller (than those of larger gas bubbles) and achieve velocities as high as 1.4 km s^{-1} during a shockwave pulse (Fong *et al* 2008). *In vitro*, high-speed images (shown in figure 4(b)) and atomic force microscopy demonstrated that microbubble-associated fluid jets resulted in a $16 \mu\text{m}$ wide pit in the surface of cells (Prentice *et al* 2005).

Within *ex vivo* tissues, the asymmetric oscillation of bubbles oscillating near boundaries demonstrated a toroidal morphology suggesting the presence of fluid jets (figure 4(c)) (Caskey *et al* 2007). With a large number of microbubbles located within a distance of tens of microns, the bubbles attract one another and fuse into a single bubble following a train of pulses (figure 4(d)).

The effect of bubble oscillation on tissue has also been examined *in vivo* by Stieger *et al* (2007). In these experiments, a chorioallantoic membrane was optically imaged using

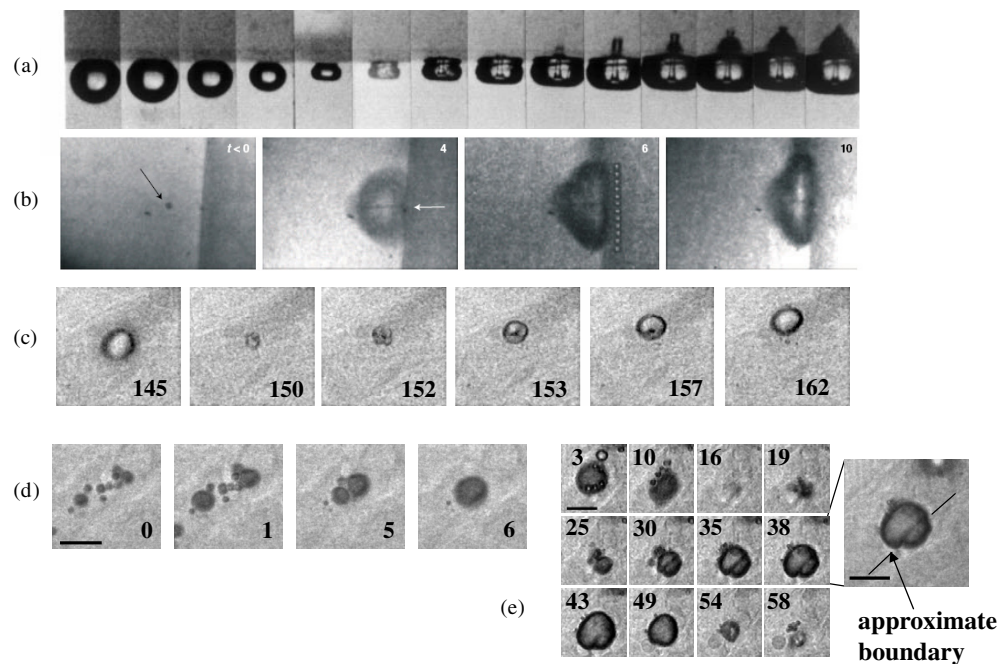


Figure 4. *In vitro* and *ex vivo* optical images of microbubbles oscillating near boundaries. (a) A cavitating bubble collapsing and impinging on a gel surface, reproduced with permission from Kodama and Tomita (2000). (b) Microbubble at boundary of cultured cell with feature in center that indicates jet formation. Reproduced with permission from Prentice *et al* (2005). (c) Microbubble oscillation in a microvessel within the rat cecum undergoing asymmetric oscillation with feature in center indicating toroidal microbubble shape. Reproduced with permission from Caskey *et al* (2007). (d) Microbubble coalescence shown in *ex vivo* microvessel with a large bubble interacting with the vessel wall. Reproduced with permission from Caskey *et al* (2007).

a fluorescently-labeled model drug during insonation with 1 and 2.25 MHz acoustic pulses at varied PRP. A sequence of images showed extravasation of the model drug during 1 MHz insonation. Electron microscopy images demonstrated disrupted vascular endothelium in a chicken embryo perfused with contrast agents, where endothelial cells with vacuoles, blebs and filipodia can be seen in the electron microscopy images acquired after applying ultrasound at transmission frequencies of 1 and 2.25 MHz (Stieger *et al* 2007).

In addition, the ultrasound pressure displaces the microbubble along the axis of the ultrasound propagation—an effect known as primary radiation force or primary Bjerknes force—shown in figure 1(a) as a downward displacement of the microbubble with each acoustic cycle (Dayton *et al* 1997, 1999a, 2002, Lum *et al* 2006, Borden *et al* 2006b). Increasing the acoustic pressure increases the radiation force and therefore increases the translational displacement. A bubble with a resting diameter equivalent to the resonant diameter for a given frequency is displaced by the greatest distance (Dayton *et al* 2002).

3. Models for microbubble dynamics

Since 1917, researchers have developed a variety of theoretical models to study gas bubble dynamics in liquids, many of which are summarized in table 2. We structure this section (and table 2) by starting with the simplest case—that of the free microbubble driven by a

Table 2. Summary of models of gas bubble dynamics.

Assumptions	Brief description	Representative references
(a) Unshelled bubble in incompressible unbounded liquid	This is the classic Rayleigh–Plesset equation assuming uniform gas pressure within the bubble, and where the gas usually obeys the polytropic law.	(Rayleigh 1917; Plesset 1949; Plesset and Prosperetti 1977; Vokurka 1985)
(b) Unconstrained small-amplitude radial oscillations	A linear approximation is applied to the Rayleigh–Plesset equation.	(Devin 1959; Fanelli <i>et al</i> 1981; Vokurka 1985; de Jong <i>et al</i> 1994a; de Jong <i>et al</i> 1992; Prosperetti 1975; Miller 1981; Allen <i>et al</i> 2001)
(c) Unconstrained small-amplitude non-spherical oscillations	Based on a linear approximation, bubble shapes, modes and scattering characteristics are investigated.	(Strasberg 1956; Gaunaurd and Uberall 1978; Sage <i>et al</i> 1979; Allen <i>et al</i> 2001; Hu <i>et al</i> 2004; Tsamopoulos and Brown 1983, 1984; Feng and Leal 1994)
Unconstrained nonlinear radial oscillations		
(d) Liquid compressibility	When the magnitude of the bubble wall velocity approaches the speed of sound in the liquid, extensions of the Rayleigh–Plesset equation are introduced that include liquid compressibility and sound radiation effects. The widely-used Gilmore equation, the Keller equation and the Herring equation fall into this category.	(Gilmore 1952; Trilling 1952; Keller and Kolodner 1956; Biasi <i>et al</i> 1972; Keller and Miksis 1980; Prosperetti 1982b; Prosperetti and Lezzi 1986; Vokurka 1986; Lezzi and Prosperetti 1987; Prosperetti 1987; Brenner 1995; Brujan 1998; Prosperetti <i>et al</i> 1988; Barber <i>et al</i> 1997; Brujan 1999; Putterman <i>et al</i> 2001; Brujan 2001; Brenner <i>et al</i> 2002)
(e) Thermal effects	Heat transfer between the bubble and liquid is included by coupling the heat diffusion equation to the modified Rayleigh–Plesset equation without adiabatic or isothermal assumptions.	(Flynn 1975a, 1975b; Kamath and Prosperetti 1989; Prosperetti 1991; Kamath <i>et al</i> 1992; Goldsztein 2004)
(f) Gas diffusion	For acoustically-driven bubble diffusion (Chomas <i>et al</i> 2001a), bubble equilibrium size is determined by the gas diffusion equation.	(Payvar 1987; Fyrrillas and Szeri 1994 1995; Lofstedt <i>et al</i> 1995)
(g) Spatial distribution of gas pressure	Gas pressure spatial distribution is coupled to bubble radial equation.	(Lin <i>et al</i> 2002; Prosperetti <i>et al</i> 1988)
(h) Shell viscosity and elasticity	The bubble shell is simplified as a layer of incompressible liquid or elastic solid. Stress and velocity continuity are met at the shell inner and outer surfaces.	(Hoff <i>et al</i> 2000; Postema and Schmitz 2007; de Jong <i>et al</i> 1994a; Glazman 1983; Frinking and de Jong 1998; Church 1995; Borden and Longo 2002; Marmottant <i>et al</i> 2005; Morgan <i>et al</i> 2000; Dayton <i>et al</i> 2001; Allen <i>et al</i> 2002)
(i) Non-Newtonian liquid	A non-Newtonian liquid model is used to describe blood flow in vessels due to the non-uniform viscosity distribution across the blood vessel cross section.	(Brujan 2001; Favelukis and Albalak 1996; Brujan 2000; Alekseev and Rybak 1999; Allen and Roy 2000b, 2000a)
(j) Translational displacement due to radiation force	Due to the pressure gradient in the liquid, a net radiation force is generated upon oscillating bubbles.	(Dayton <i>et al</i> 2002; Zhao <i>et al</i> 2004; Pelekasis <i>et al</i> 2004; Doinikov and Dayton 2006)

Table 2. (Continued.)

Assumptions	Brief description	Representative references
Constrained nonlinear oscillations		
(k) Near a surface	Both analytical solutions and numerical simulations have been developed. Image theory and boundary integral methods are utilized	(Oguz and Prosperetti 1990a, 1990b, 1993, Takahira 1997, Yuan and Prosperetti 1997, Zeff <i>et al</i> 2000, Brujan <i>et al</i> 2001, 2002, 2004, Fong <i>et al</i> 2006, Cui <i>et al</i> 2006, Krasovitski and Kimmel 2004, Nyborg 1958, Sato <i>et al</i> 1994, Afanasiev and Grigorieva 2006, Blake <i>et al</i> 1999, Bremond <i>et al</i> 2006)
(l) Within a tube	During insonation, ultrasound contrast microbubbles oscillate within blood vessels. To study bubble oscillation within a small vessel, the vessel can be simplified as a rigid or compliant tube	(Yuan <i>et al</i> 1999, Ory <i>et al</i> 2000, Hu <i>et al</i> 2005, Qin and Ferrara 2006, Qin <i>et al</i> 2006, Ye and Bull 2006, Qin and Ferrara 2007)
(Unconstrained nonlinear and constrained linear) non-spherical oscillations		
(m) Unconstrained nonlinear non-spherical oscillations	The assumption of the spherical morphology of the bubble–liquid interface is removed. The bubble–liquid interface shape is determined either by gas–liquid interaction or by a predetermined shape profile	(Crum 1979, Klaseboer <i>et al</i> 2006, Yang <i>et al</i> 1993, Gordillo and Fontelos 2007, Eggers <i>et al</i> 2007)
(n) Constrained small-amplitude non-spherical oscillations	Linear approximation is used and the boundary conditions are enforced	(Sassaroli and Hynnen 2004, 2005, Oguz and Prosperetti 1998, Cui <i>et al</i> 2006)
Bubble–bubble interaction		
(o) Unconstrained bubble cloud or bubble–bubble interaction dynamics	The bubbly liquid is described by the average governing equations	(Lu <i>et al</i> 1990, Pelekasis <i>et al</i> 2004, Yoon <i>et al</i> 1991, Commander and Prosperetti 1989, Allen <i>et al</i> 2003, Leroy <i>et al</i> 2005, Druzhinin <i>et al</i> 1996, Karpov <i>et al</i> 2003, Sarkar and Prosperetti 1993)
(p) Constrained bubble cloud or bubble–bubble interaction dynamics	Image theory is utilized to solve the modified Rayleigh–Plesset equations	(Payne <i>et al</i> 2005, Bremond <i>et al</i> 2006, Hamilton <i>et al</i> 2005)

low amplitude sound field in an infinite fluid. Clearly, there are many levels of assumptions required to apply this model to the real application—that of encapsulated bubbles driven by high amplitude sound fields and traveling within small blood vessels which contain a high concentration of red blood cells. In order to accurately model the behavior of contrast microbubbles *in vivo*, a series of increasingly complex models has been developed, still typically including a number of assumptions and simplifications in each case. For example, recent treatments of the behavior of bubbles within a bubble cloud have begun with the unshelled bubble (Hamilton *et al* 2005), while simultaneously others address the behavior of a single microbubble oscillating non-spherically within a small vessel (Qin and Ferrara 2006).

The fundamental equations of bubble dynamics were developed by Rayleigh and Plesset (Rayleigh 1917, Vokurka 1985, Plesset and Prosperetti 1977, Plesset 1949) (table 2(a)), neglecting liquid compressibility effects and assuming that the gas pressure in the bubble is uniform and obeys the polytropic law, which is given by

$$\rho \left(\ddot{R}R + \frac{3}{2} \dot{R}^2 \right) = p_g(t) - p_0 - p_i(t) - \frac{4\eta\dot{R}}{R} - \frac{2\sigma}{R}, \quad (1)$$

where R_0 is the bubble radius at equilibrium, $\dot{R}$ and $\ddot{R}$ represent respectively the first- and second-order time derivatives of the bubble radius R , p_0 is the hydrostatic pressure, $p_i(t)$ is the incident ultrasound pressure in the liquid at an infinite distance, $p_g(t)$ is the uniform gas pressure within the bubble, and ρ , σ and η are the density, surface tension and viscosity of the bulk fluid, respectively. A historical review of the development of this equation was given by Plesset and Prosperetti (1977). The gas pressure within the bubble depends on the volumetric change with respect to its equilibrium state and heat diffusion across the bubble wall (Prosperetti *et al* 1988). The polytropic law neglects heat diffusion and relates gas pressure and bubble volume as given by

$$p_g(t) = \left(p_0 + \frac{2\sigma}{R_0} \right) \left(\frac{R_0}{R} \right)^{3\kappa}, \quad (2)$$

where κ is the polytropic exponent and R_0 is the bubble equilibrium radius. If the thermal diffusion length in the gas is greater than the bubble radius, the bubble will behave isothermally (i.e. $\kappa \approx 1$). However, if the thermal diffusion length in the gas is much smaller than the bubble radius and the bubble radius is much less than the wavelength of sound in the bubble, the bubble will behave adiabatically (i.e. $\kappa \approx \gamma$, the specific heat ratio of the gas within the bubble) (Prosperetti 1982a).

Using a small-amplitude oscillation assumption, the Rayleigh–Plesset equation has been widely applied to study many aspects of bubble dynamics such as bubble natural frequency, acoustic scattering characteristics and thermal damping effects (Devin 1959, Fanelli *et al* 1981, Vokurka 1985, de Jong *et al* 1994b, 1992, Prosperetti 1975, Miller 1981, Strasberg 1956, Gaunard and Uberall 1978, Sage *et al* 1979, Allen *et al* 2001, Hu *et al* 2004, Tsamopoulos and Brown 1983, 1984, Feng and Leal 1994) (table 2(b) and (c)). Taking into account the effect of the surface tension, the Minnaert expression for bubble resonance frequency can be given by (Minnaert 1933, Miller 1981)

$$\omega_0 = \frac{1}{R_0} \left[\frac{3\kappa}{\rho} \left(p_0 + \frac{2\sigma}{R_0} \right) - \frac{2\sigma}{\rho R_0} \right]^{1/2}. \quad (3)$$

The emitted ultrasound pressure at distance r from the bubble center is

$$p(r) = \frac{\rho R}{r} (2\dot{R}^2 + R\ddot{R}). \quad (4)$$

As verified by numerous experiments, the Rayleigh–Plesset equation works well when the Mach number (the ratio of the velocity of the bubble wall to the sound speed in the liquid, $\dot{R}/c$) is small (Putterman *et al* 2001). However, when the incident pressure amplitude increases, the Mach number approaches unity and sound radiation and liquid compressibility become important. For higher pressure amplitudes and larger radial oscillation, extensions of the Rayleigh–Plesset equation have been proposed, including the Keller equation, the Herring equation and the Gilmore equation (Gilmore 1952, Trilling 1952, Keller and Kolodner 1956, Keller and Miksis 1980, Prosperetti and Lezzi 1986, Lezzi and Prosperetti 1987, Prosperetti 1987, Prosperetti *et al* 1988, Brenner 1995, Lofstedt *et al* 1995, Barber *et al* 1997) (table 2(d)). Prosperetti *et al* (1988) and Prosperetti and Lezzi (1986) demonstrated that there is a one-parameter family of equations to describe bubble oscillation in unbounded liquids, namely

$$\begin{aligned} & \rho \left\{ \left[1 - (\lambda + 1) \frac{\dot{R}}{c} \right] \ddot{R}R + \frac{3}{2} \dot{R}^2 \left[1 - \left(\lambda + \frac{1}{3} \right) \frac{\dot{R}}{c} \right] \right\} \\ &= \left[1 + (1 - \lambda) \frac{\dot{R}}{c} \right] \left[p_B(t) - p_0 - p_i \left(t + \frac{R}{c} \right) \right] + \frac{R}{c} \dot{p}_B(t), \end{aligned} \quad (5)$$

where λ is an arbitrary parameter that preserves the first order of accuracy of the equation and $p_B(t)$ is the liquid pressure on the external side of the bubble wall, which depends on the internal bubble gas pressure on the wall

$$p_B(t) = p_g(t) - \frac{4\eta\dot{R}}{R} - \frac{2\sigma}{R}. \quad (6)$$

Setting the parameter λ equal to 0 recovers the Keller equation (Keller and Kolodner 1956, Keller and Miksis 1980), and λ equal to 1 results in the Herring and Trilling equation (Trilling 1952). The effects of other factors such as non-uniform pressure within the bubble and the heat and gas transfer between the bubble and liquid have also been extensively examined (table 2(e), (f) and (g)). The Keller-like equation written in terms of the enthalpy has been verified by extensive experimental results and demonstrated to yield results that are in agreement with full partial differential equation numerical simulations (Prosperetti and Lezzi 1986, Lezzi and Prosperetti 1987, Lin *et al* 2002).

Ultrasound contrast agents typically have a shell with a thickness from 10 to 100 nm, motivating studies of the effects of bubble shell viscosity and elasticity (table 2(h)). Roy and his co-workers first treated the bubble shell as a simple viscous liquid and found good agreement between model predictions and *in vitro* experimental measurements of the cavitation threshold (Roy *et al* 1990). De Jong and his co-workers later (de Jong *et al* 1994a, 1994b, de Jong and Hoff 1993, de Jong *et al* 1992) modeled bubble shells as layers of elastic solids and studied acoustic attenuation and backscatter and nonlinear oscillation, validated by experimental results.

Models that include shell properties using classical mechanical principals have been developed. Church (1995) derived a Rayleigh–Plesset-like equation describing the dynamics of encapsulated gas bubbles, assuming that coating material is a layer of an incompressible, viscous-elastic solid. Hoff *et al* (2000) then developed a model using viscous and elastic properties of the shell to describe polymeric microbubble behavior. Modified Rayleigh–Plesset equations have been developed for thin and thick viscoelastic-shelled agents, examining shell viscosity and elasticity effects (Allen *et al* 2002, Morgan *et al* 2000). These equations were validated by direct comparison of the predicted bubble radius with optically-measured streak images (Allen *et al* 2002, Morgan *et al* 2000). For a gas bubble encapsulated by a layer of viscous-elastic solid shell or viscous liquid shell, the Rayleigh–Plesset-like equations derived by Church (1995) and Allen *et al* (2002) can be written in a simple formula as

$$\begin{aligned} \ddot{R}_1 R_1 \left[1 + \left(\frac{\rho - \rho_s}{\rho_s} \right) \frac{R_1}{R_2} \right] + \dot{R}_1^2 \left[\frac{3}{2} + \frac{\rho - \rho_s}{\rho_s} \frac{4R_2^3 - R_1^3}{2R_2^3} \frac{R_1}{R_2} \right] \\ = \frac{1}{\rho} \left\{ p_g(t) - p_0 - p_i(t) - \frac{4\eta R_1^2 \dot{R}_1}{R_2^3} \right. \\ \left. - \frac{2(\sigma_1 + \sigma_2)}{R} - 4 \frac{R_2^3 - R_1^3}{R_2^3 R_1} [\mu_s \dot{R}_1 + G_s (R_1 - R_{10})] \right\}, \end{aligned} \quad (7)$$

where R_1 is the inner radius of the agent, R_2 is the outer radius, R_{10} is the inner radius at equilibrium, σ_1 is the surface tension at the inner radius, σ_2 is the surface tension at the outer radius and ρ_s is the shell density. G_s equals the Lamé constant (the modulus of rigidity) if the agent is an elastic solid shell (Church 1995) and G_s equals 0 for a viscous liquid shell (Allen *et al* 2002).

While the above discussion demonstrates that substantial effort has been applied to the development of models for shelled microbubbles, accurately representing the shell properties before and after insonation continues to be a challenging problem. Given the small thickness of the shell (typically nanometers) and small diameter of the bubble (typically microns),

evaluating the mechanical properties of the shell material *in situ* is challenging. Moreover, the molecular scale properties of lipid membranes cannot be simply incorporated within models that describe only the radial component of oscillation. Lipid molecules self-assemble at the gas–liquid interface. When the lipid membrane is compressed (as might be expected during the compressional half-cycle), the monolayer can reversibly buckle or shed lipid into the solution (Ridsdale *et al* 2001). During rarefaction, the lipid membrane expands, potentially leaving patches of uncoated gas–liquid interface. Classical mechanics cannot fully characterize these scenarios and therefore specialized approaches for each shell material are required. Further, in order to translate these models to a blood and tissue environment, investigations of the effect of tissue elasticity and non-uniform blood viscosity on oscillations of ultrasound contrast agents have been undertaken and efforts have also been made to develop theoretical models for radial oscillations of gas bubbles in non-Newtonian liquids (table 2(i)).

After intravenous bolus injection, low amplitude ultrasound pulses can be administered to deflect the drug-coated bubbles toward the blood vessel wall and facilitate imaging or drug delivery. A model was developed to predict the radial oscillation and simultaneous translational displacement of an encapsulated gas bubble after insonation by clinically applicable megahertz ultrasound pulses (Dayton *et al* 2002, Zhao *et al* 2004) (table 2(j)). As with the free bubble, the translational displacement of the shelled bubble was shown to be determined by the combination of the driving force, the translational added mass, the oscillatory added mass and the quasistatic drag force.

Oscillations of ultrasound contrast agents within the blood pool are constrained by blood vessels, and this effect is greatest within small vessels. The boundary can be simplified and image theory or boundary integral methods are then applied in order to calculate bubble oscillations (table 2(k)). This simplification of using image theory works well when the ratio of bubble diameter to the tube diameter is small. Alternatively, a bubble oscillating within a small tube has been described by analytical approximations (Hu *et al* 2005, Yuan *et al* 1999) or by numerical computations (Ory *et al* 2000, Qin and Ferrara 2006, 2007, Qin *et al* 2006, Ye and Bull 2006) (table 2(l)). Our investigation has demonstrated that oscillations of a bubble within a small vessel depend on the vessel and bubble diameter and mechanical properties of vessel and connecting tissues (Qin and Ferrara 2006, 2007). The natural frequency of the bubble within a small rigid vessel is substantially decreased with decreasing vessel size. However, for the same-sized vessels, the natural frequency of a bubble increases with decreasing vessel rigidity (Qin and Ferrara 2007). For a bubble constrained in a small vessel, the non-radial component of oscillation is significant (Hu *et al* 2005, Qin and Ferrara 2006) (figure 4) and should be included as the effect of the oscillation on the vessel wall is evaluated. With a transmitted PRP of 0.5 MPa and a center frequency of 1 MHz, the expansion of a microbubble (with an initial diameter of 3 μm) within a compliant vessel (of diameter 8 μm) can dilate the vessel wall by a distance as large as a few microns (Qin and Ferrara 2006) (figure 5).

In drug delivery applications, the ultrasound pressure ($\text{PRP} > 0.5 \text{ MPa}$) can be substantially larger than the pressure used in traditional imaging ($\text{PRP} \sim 50 \text{ kPa}$). When insonified by a high ultrasound pressure, bubbles become unstable, rapidly collapse and fragment; thus, models for stability and mode analysis are summarized in table 2(m) and (n). Further, given that drug delivery studies often use high microbubble concentrations, models of bubble–bubble interaction, bubble–bubble–vessel interaction and bubble cloud/density effects have been developed (table 2(o) and (p)), often requiring substantial computational resources to solve partial differential equations. In limited cases, solutions have been obtained assuming a uniform formulation or using image theory.

In summary, understanding and accurately predicting microbubble oscillation *in vivo* is a challenging problem. While a great deal of progress has been made in accurately representing

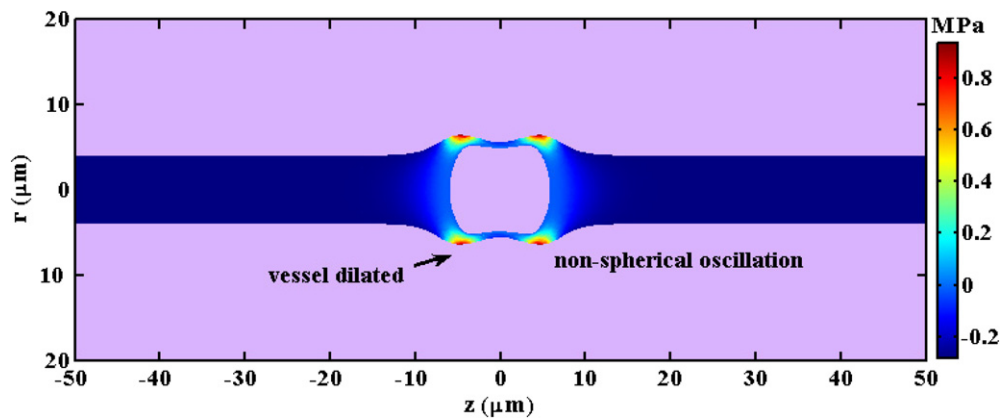


Figure 5. A snapshot of the predicted pressure field induced by a microbubble with an initial diameter of $3\ \mu\text{m}$ as it oscillates in a microvessel with an inner diameter of $8\ \mu\text{m}$ in an ultrasound field with a PRP of $0.5\ \text{MPa}$ and center frequency of $1\ \text{MHz}$.

the radial oscillation of the shelled microbubble in an infinite fluid, still refinements and improvements of such models continue, e.g. Doinikov and Dayton (2006). Moreover, modeling of nonlinear and aspherical oscillation, modeling of bubble clouds and modeling of bubble interaction with neighboring cells and vessel walls require additional development. Therefore, gaining a perspective on the hierarchy of approaches to this problem (as in table 2) can be of value.

4. Microbubble-based imaging techniques

Echoes from microbubbles change in response to the frequency, amplitude or phase of the ultrasound pulse as a result of changes in wall velocity and acceleration, as well as microbubble destruction. The design of microbubble imaging strategies (table 3 and figure 6) not only includes the choice of the center frequency, bandwidth, pressure and phase of each individual pulse but also involves the design of a pulse train incorporating multiple pulses for which these parameters are altered between pulses, in order to facilitate the discrimination of microbubble and tissue echoes.

Early contrast agent imaging techniques were based on the detection of the harmonics returned from each pulse (Schrope and Newhouse 1993, Villanueva and Kaul 1995). With single pulse detection techniques, each echo is filtered as it is received in order to remove frequency components outside the desired harmonic or subharmonic band (Shankar *et al* 1998, Shi *et al* 1999, Chomas *et al* 2002, Forsberg *et al* 2000, Cheung *et al* 2008). Such filtering of the radiofrequency pulse is also known as fast time processing, as each pulse is processed as it is received (figure 6). For harmonic and subharmonic strategies, a narrowband pulse is typically transmitted in order to minimize the spectral overlap between the transmitted and received pulses. Subharmonic strategies can employ the transmission of a pulse centered at the resonance frequency of the microbubble or at a multiple of the resonance frequency (Chomas *et al* 2002). Transmission of a pulse centered at twice the resonance frequency results in resonant oscillation of the microbubble (and thus an apparent subharmonic), even with a low transmission pressure.

Alternatively, transmission of a short (single cycle) pulse, with a sufficiently high pressure ($>250\ \text{kPa}$) and with rarefaction preceding compression, results in a rapid microbubble

Table 3. Microbubble contrast agent imaging strategies.

	Transmitted pulse	Radiofrequency (fast-time) filter	Pulse-to-pulse (slow-time) filter	Reference
Single pulse strategies				
Harmonic (narrowband)	Multicycle pulse with a center frequency near the microbubble resonance and low amplitude	High pass	N/A	(Villanueva and Kaul 1995)
Transient (wideband)	Single cycle pulse with a center frequency of ~2 MHz and PRP of ~200–500 kPa	High pass	N/A	(Kruse and Ferrara 2005)
Subharmonic	Multicycle pulse with a center frequency either near the microbubble resonance or a multiple of resonance	Low pass	N/A	(Cheung <i>et al</i> 2008, Chomas <i>et al</i> 2002, Forsberg <i>et al</i> 2000, Shankar <i>et al</i> 1998, Shi <i>et al</i> 1999)
Coded excitation	Many possible codes	Many possible filtering methods	Can be extended to multipulse strategy	(Borsboom <i>et al</i> 2004, 2005, Gong <i>et al</i> 2007, Sun <i>et al</i> 2007)
Low-power multipulse strategies				
Harmonic power Doppler	Sequential pulses as in the harmonics above	High pass	High pass (difference between returns)	(Senior <i>et al</i> 2000, Simpson <i>et al</i> 2001, Bauer <i>et al</i> 1999, Chang <i>et al</i> 1995)
Phase inversion	Sequential harmonic or transient pulses with phase shift between transmission	Often high pass but not required	Low pass (sum of returned echoes)	(Graubner <i>et al</i> 1997, Burns <i>et al</i> 2000, Harvey <i>et al</i> 2000a, 2000b, Morgan <i>et al</i> 1998)
Dual frequency-radial modulation	Sequential or simultaneous transmission of harmonic pulses with varied center frequency	Low pass		(Chen <i>et al</i> 2006b, Masoy <i>et al</i> 2008, Cherin <i>et al</i> 2008)
Power modulation	Sequential pulses differing in pressure amplitude	Not required	Scale and subtract	(Porter <i>et al</i> 2003, Brock-Fisher and Prater 2006)
CPS	Sequential pulses differing in pressure and phase	Not required	Low pass (sum of returned echoes)	(Phillips and Gardner 2004, Solbiati <i>et al</i> 2004, Stieger <i>et al</i> 2008).
High-power multipulse strategies				
Destruction–reperfusion	Train of similar pulses is transmitted	Varied	Loss of correlation or changes in echo amplitude are assessed	(Wei <i>et al</i> 1998, 2001a, 2001b, Chomas <i>et al</i> 2003, Krix <i>et al</i> 2004, Potdevin <i>et al</i> 2004, Dijkmans <i>et al</i> 2004)

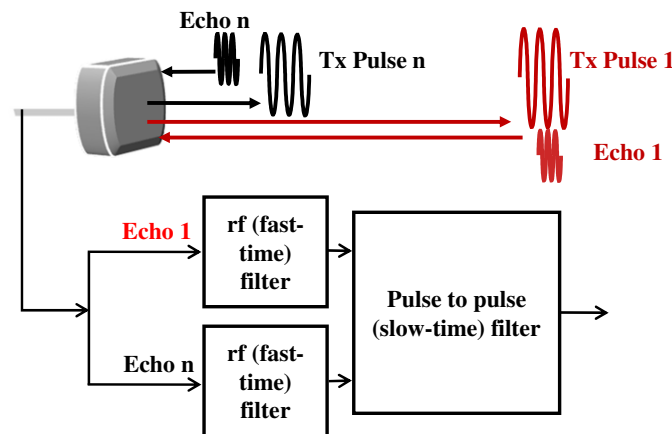


Figure 6. Diagram of the generation and processing of contrast agent echoes for the creation of images, corresponding to the entries in table 3. The detection of echoes from microbubble contrast agents may be accomplished by processing the returned echo from a single pulse or the returned echoes from a train of ultrasound pulses.

collapse and wideband echo and can be used for single-cycle ‘transient’ contrast imaging (Kruse and Ferrara 2005). Coded pulse transmissions involving linear or nonlinear frequency modulation have also been explored, where the differences in the received echoes resulting from microbubbles and tissue are exploited to further improve the signal-to-noise ratio (Borsboom *et al* 2003, 2004, 2005, Gong *et al* 2007, Sun *et al* 2007). Mixing low frequency and high frequency transmission to modulate the diameter of the microbubble has also been explored (Bouakaz *et al* 2007, Chen *et al* 2006b, Masoy *et al* 2008, Cherin *et al* 2008).

When multiple pulses are transmitted and received, each pulse can be filtered to tailor the received bandwidth, again removing frequency components that correspond to frequencies within the transmitted pulse. In addition, pulse-to-pulse filtering is applied to the echoes corresponding to a fixed depth, in order to further improve the discrimination of tissue and bubble echoes. This filter is typically called a slow-time filter. Following the transmission of a train of identical pulses, a high-pass pulse-to-pulse filter can reject echoes from fixed tissue and increase the sensitivity to moving microbubbles; such processing is typically known as power Doppler processing. Combining fast-time and slow-time high-pass filters produces harmonic power Doppler processing, frequently used for perfusion imaging (Senior *et al* 2000, Simpson *et al* 2001, Bauer *et al* 1999, Chang *et al* 1995). Disadvantages of this technique include limitations on the detection of capillary perfusion due to the very slow capillary flow.

Low-amplitude pulse trains used in contrast imaging involve the transmission of pulses with altered phase (Graubner *et al* 1997, Burns *et al* 2000, Harvey *et al* 2000a, 2000b, Morgan *et al* 1998), amplitude (Porter *et al* 2003, Brock-Fisher and Prater 2006) or both phase and amplitude (Phillips and Gardner 2004, Solbiati *et al* 2004, Stieger *et al* 2008). Summation of the returned echoes cancels the return from stationary tissue, while again preserving sensitivity to microbubble echoes. An advantage of the use of such pulse trains is the elimination of the requirement for the radiofrequency (fast-time) filter. Given the limited bandwidth of current transducers, elimination of this requirement can potentially increase the signal-to-noise ratio of the measurement (effectively using the entire received bandwidth). An example of the gray-scale image of a murine tumor is presented in figure 7(a), where the tumor is circled in red, with a hypoechoic gray scale. An image of the contrast agent echoes, estimated with the contrast pulse sequence (CPS) developed by Siemens Medical Solutions (MountainView,

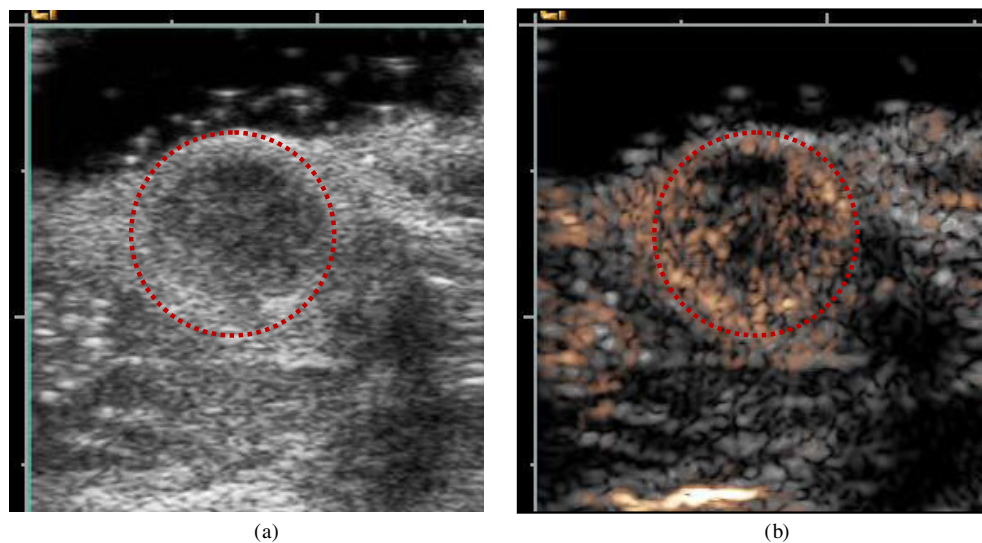


Figure 7. Examples of ultrasound contrast agent images of Met-1 murine tumor. (a) B-mode and (b) corresponding contrast pulse sequence (CPS) (gold color) image overlaid on B-mode (gray). Region of low echogenicity in the B-mode image (circled in red) is the region of tumor, with vascular density shown by the gold contrast agent overlay.

CA), can then be overlaid on the gray-scale image (figure 7(b)). The spatial variation in contrast agent density in such images correlates with the tumor morphology and density of viable tumor cells, as estimated during histology using hematoxylin and eosin (Chomas *et al* 2003, Broumas *et al* 2005).

The blood flow rate is estimated in contrast ultrasound by combining high-amplitude pulse transmissions, designed to destroy the contrast agent, with pulses designed to estimate the replenishment of microbubbles into the region of interest (Wei *et al* 1998, 2001a, 2001b, Chomas *et al* 2003, Krix *et al* 2004, Potdevin *et al* 2004, Dijkmans *et al* 2004). A parametric image of the time required for contrast replenishment within a region can then be created, noting that the flow rate will vary across the image (Chomas *et al* 2003). Each color-encoded pixel represents an estimate of the contrast replenishment rate for a small region. Depending on the center frequency of the ultrasound imaging pulses, the spatial resolution of each estimate is on the order of hundreds of microns to millimeters. Within cancerous tumors, the flow of the contrast agent is not always continuous. Therefore, estimating the variations in flow rate with time can be of value (Chomas *et al* 2003). Chomas *et al* color-coded the regions that did not maintain a consistent flow rate over time with a distinct color marker (red).

4.1. Imaging applications

Microbubble contrast imaging has found many applications in cardiology, including the assessment of wall motion abnormalities and perfusion defects, with the clinical aspects described recently (Kaufmann *et al* 2007, Kaul 2008). Radiology applications focus on cancer and peripheral vascular disease, where the estimation of microvascular density and flow rate is particularly important. Ultrasound is an ideal technique to use in serial studies of drug therapies, e.g. anti-angiogenic treatment (Tang *et al* 2003). Although contrast-enhanced gray-scale imaging techniques (such as pulse inversion, power modulation and CPS) are widely used and the use of ultrasound to monitor drug therapies is important (Lucidarme

et al 2004, Forsberg *et al* 2002), quantitative measures are not yet validated for clinical studies but are currently of great interest. The development of an accepted ultrasound measurement of vascularity or flow rate would be of great value for basic science and clinical studies.

4.2. Targeted, lipid-shelled microbubbles and their detection

Lipid-shelled microbubble stability is enhanced by polymer-grafted lipids, with concentrations of 5–9 molar percent of polyethylene glycol typically used. Targeting ligands are then attached to the distal end of a polymer either before or after microbubble generation. Large proteins are typically attached after the contrast agent is formed due to the increased temperature and agitation used in the formation of the microbubble; peptides and other small molecules can be attached beforehand. A large number of ligands are typically required to ensure adequate coverage, given the large (micron diameter) size of the microbubble. A biotin–streptavidin linkage is used most commonly to attach the targeting ligand to the lipid shell (Klibanov 2005, Lanza and Wickline 2001).

Klibanov *et al* originally demonstrated the *in vitro* use of targeted microbubbles by studying microbubbles adherent to avidin-coated petri dishes (Klibanov *et al* 1997, 1998). More recently, targeted agents have been shown to successfully detect inflammation, transplant rejection and atherosclerosis (Nakashima *et al* 1998, Jian-Guo *et al* 2004, Lindner *et al* 2001, Weller *et al* 2003). Angiogenesis has been the most common application for targeted contrast agents with peptide and antibody-based strategies reported, as well as dual ligand techniques reported (Ellegala *et al* 2003, Stieger *et al* 2008, Weller *et al* 2005, Lee *et al* 2008, Willmann *et al* 2008).

Most reports of imaging with targeted microbubbles have involved observation of the entrance of the microbubbles into the field of view followed by an assessment of the image intensity over time, culminating with a destructive pulse. The signal amplitude following the destructive pulse is assumed to represent circulating agents and the signal before the destructive pulse is assumed to include circulating plus bound agents (Ellegala *et al* 2003, Lindner *et al* 2001, Stieger *et al* 2008). The development of new techniques to detect targeted agents is an area of intense study where radiation force and changes in spectral characteristics can enhance the detection of bound bubbles.

5. Effect of contrast agents on tissue/vasculature

The mechanism for the vascular permeability enhancement observed with ultrasound contrast agents is a subject of active research by multidisciplinary groups, including physicists, engineers and biologists. Here, common acoustic parameters and bubble properties known to enhance permeability are examined while considering the *in vivo* environment of the bubble. Changes in microbubble oscillation due to insonation frequency, PRP, pulse duration, pulse repetition frequency, contrast agent dosage, size and shell properties are each considered.

5.1. Targets and barriers for delivery with ultrasound contrast

Regardless of function, a drug must overcome many of the body's natural barriers in order to be effective. Ultrasound contrast agents circulate for an interval that is typically on the order of minutes. Once bubbles have reached the target site, ultrasound can be applied to locally increase permeability. Therefore, sufficient contrast agent density and an acoustic window from the transducer to the target site are two requirements for drug delivery with ultrasound contrast agents. The brain, heart, pancreas and many thrombi and cancerous tumors meet

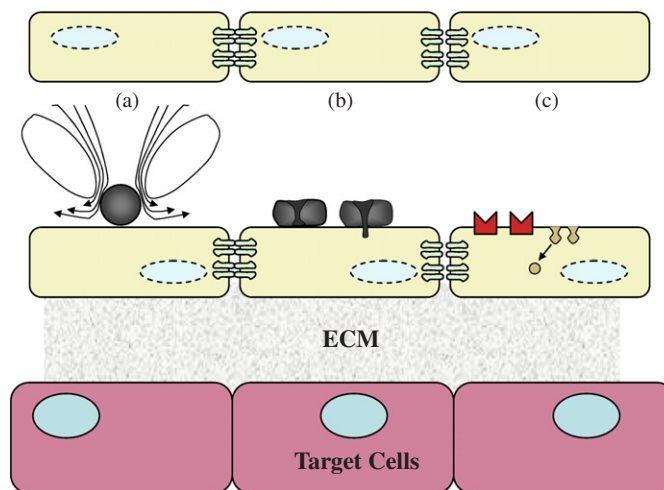


Figure 8. Hypothesized mechanisms of drug transport across endothelium. Illustration of (a) local shear stress created on cell during microbubble oscillation, (b) fluid jet formation and (c) intra-cellular transport that are hypothesized to result from the stresses induced by microbubble activity, including generation of gaps at tight junctions, expression of cell adhesion molecules due to inflammatory process and the creation of vesicles for trans-cellular transport.

these requirements and have been targets of ultrasound-enhanced delivery strategies (Choi *et al* 2007, Bekerredjian *et al* 2003, Chen *et al* 2006a, Tsutsui *et al* 2006, Tartis *et al* 2006).

In an idealized vascular network, blood flows successively through large arteries, small arteries, arterioles, capillaries, post-capillary venules, venules, small veins and large veins (Berne *et al* 2004). Exchange of molecules from the blood to organs generally occurs in the capillaries and post-capillary venules (diameters of approximately 4–8 μm). Once in these small vessels, a molecule must move beyond the vascular endothelium into the interstitial space for successful delivery to the organ of interest (illustrated in figure 8). Capillaries can be grouped into three types in the human body: continuous capillaries, fenestrated capillaries and sinusoids. A continuous capillary has immediately adjacent endothelial cells that create a continuous and uniform barrier overlaid on the basement membrane. These are typically found in skin, connective tissues, striated and smooth muscle, lung and brain. Fenestrated capillaries are found in the renal glomeruli, the intestinal villi, endocrine glands and the pancreas. The cytoplasm of cells on each side of the nuclear region is thin (40–60 nm) and is perforated at intervals by a system of ‘pores’ which vary greatly in number and size (their diameters may range from 30 to 100 nm). It is possible that the presence of pores in fenestrated capillaries may make endocrine glands and kidneys more susceptible to mechanical effects of contrast agents, whereas continuous capillaries surrounded by connective tissue may present a more robust boundary. The sinusoids are found in tissues such as bone marrow, spleen, liver, suprarenal/parathyroid glands, carotid and coccygeal bodies. Sinusoids tend to be larger (often larger than 10 μm), have a wider, irregular lumen and very thin wall and can be continuous or fenestrated (Strandring 2005). The diameter and composition of the surrounding vessel affect microbubble oscillation amplitude and frequency (where oscillation was described in section 3 of this review) and therefore the differences between capillary types are expected to alter the threshold for biological effects.

Another important variable for delivery is the composition of the interstitial space in the organ. Fung *et al* have suggested that capillaries derive their structure from the surrounding

organ, so that the capillaries can be viewed as a tunnel in a gel (Fung 1993). Considering the wide variation of Young's modulus values in tissues, one might expect tissues with higher elastic modulus, such as muscles, to present a more rigid boundary beyond the endothelial cells than an organ with a lower Young's modulus, such as the kidneys. *In vitro* studies support this hypothesis—rigid PMMA tubes with a diameter of 12 μm significantly constrain microbubble oscillation, while oscillation is less constrained in *ex vivo* vessels with a smaller rigidity than PMMA tubes (Caskey *et al* 2006, 2007).

5.2. Important parameters and their impact on delivery and biological effects

5.2.1. Transmission frequency and PRP. Mechanical index (MI), a commonly displayed parameter on ultrasound systems, is defined as the derated PRP in megaPascals (at the point of the maximum pulse intensity integral) divided by the square root of the ultrasonic center frequency in megahertz. The upper limit for MI was chosen to be consistent with the acoustic output produced by ultrasound systems prior to 1976, where a guideline for MI in the absence of a microbubble contrast agent is 1.9 (Meltzer 1996). However, MI was created prior to the widespread use of microbubble contrast agents and therefore did not envelope the full parameter space and mechanisms of biological effects associated with these agents.

The threshold for cavitation has long been known to be substantially reduced in the presence of gas bodies; the MI threshold for hemorrhage in mouse lung was observed to be 0.4 (Frizzell *et al* 1994). More recently, Church and O'Brien proposed a new index for acoustic output for imaging near the lungs, including pulse duration and pulse repetition frequency, in addition to the parameters described in the traditional MI (Church and O'Brien 2007).

Defining a relationship between acoustic pressure and center frequency that is predictive of biological effects in the presence of contrast agents is under study, with several recent papers proposing the ratio of acoustic pressure and center frequency (Qin and Ferrara 2006, Miller *et al* 2008). A limited number of studies have used clinical ultrasound systems or relevant parameters together with a clinically relevant ultrasound dose. One such recent paper by Miller *et al* examined the dependence of frequency and PRP on the occurrence of glomerular capillary hemorrhage (GCH), demonstrating that constant values of the ratio of PRP to the transmitted center frequency yielded a consistent effect. A threshold of 0.5 (for PRP/frequency) was predictive of hemorrhage in the rat kidney (figures 9(a) and (b)) (Miller *et al* 2008). Using 1 and 2.25 MHz, Stieger *et al* reported that the occurrence of extravasation of FITC-labeled dye in the chorioallantoic membrane model demonstrated a similar dependence (on PRP/frequency), with an apparent threshold of approximately 0.6 (Stieger *et al* 2007). These observations are supported by theoretical simulations that indicate that the frequency dependence of circumferential stress on a small compliant vessel scales with the ratio of PRP to center frequency, rather than MI (Qin and Ferrara 2006). The absolute threshold value (for the PRP/frequency ratio) may be greater in human studies due to microbubble-filled intervening tissue and species and tissue-dependent microvascular characteristics. However, the relationship between pressure, frequency and circumferential stress is generally applicable.

Surveying the microbubble-induced bioeffects and gene delivery literature, without regard to injected dose and pulse length, yields a broader range of the dependence of observed biological effects on PRP and center frequency (figure 9(c)). Many of the studies summarized in figure 9(c) used high dosages of the contrast agent Optison and high pulse duty cycles and therefore one must carefully assess the dosage and ultrasound parameters used in each study. Here, subsequent sections discuss pulsing parameters and dosage and contrast agent type.

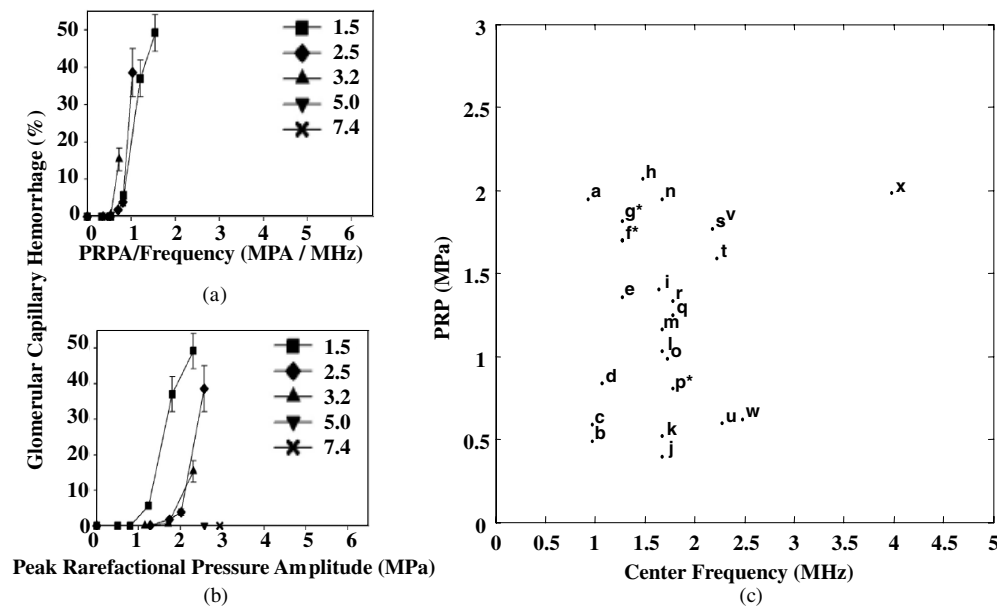


Figure 9. (a) Glomerular capillary hemorrhage versus the ratio of PRP to center frequency for a range of transmitted ultrasound frequencies using a diagnostic ultrasound scanner to insonify the rat kidney. Reproduced with permission from Miller *et al* (2008). (b) Glomerular capillary hemorrhage versus PRP for a range of transmitted ultrasound frequencies using a diagnostic ultrasound scanner to insonify the rat kidney. For (a) and (b) pulses with 1–3 cycles were generated with a diagnostic ultrasound system at the specified frequencies. Reproduced with permission from Miller *et al* (2008). (c) Survey of transmission frequencies and PRP observed in the literature. Asterisk indicates multiple studies using same parameter. a: (Lawrie *et al* 2000), b: (Stieger *et al* 2007), c: (Endoh *et al* 2002), d: (Miller and Gies 1998), e: (Chen *et al* 2002), f*: (Shohet *et al* 2000, Vannan *et al* 2002, Bekeredjian *et al* 2003, 2005, Guo *et al* 2004), g*: (Frenkel *et al* 2002, Chen *et al* 2003), h: (Miller *et al* 2005), i: (Van Der Wouw *et al* 2000), j: (Li *et al* 2004), k: (Li *et al* 2003), l: (Chapman *et al* 2005), m: (Pislaru *et al* 2003), n: (Kobayashi *et al* 2003), o: (Christiansen *et al* 2003), p*: (Kobayashi *et al* 2003, Kobayashi *et al* 2002), q: (Wible *et al* 2002), r: (Ay *et al* 2001), s: (Teupe *et al* 2002), t: (Stieger *et al* 2007), u: (Skyba *et al* 1998), v: (Miller and Gies 2000), w: (Miller and Qudus 2000), x: (Wible *et al* 2002).

5.2.2. Pulse duration/duty cycle. The threshold for microbubble fragmentation has been described by Chomas *et al* (Chomas *et al* 2001b). However, secondary radiation forces produced by the ultrasound wave facilitate fusion of the fragmented microbubble during continuous insonation. Thus, the microbubble can repeatedly fragment and re-combine during a pulse and is more likely to remain as a set of discrete fragments after or between pulses. Therefore, pulse duration and duty cycle are important parameters for drug delivery sequences and the induction of biological effects; however, these parameters cannot typically be independently varied on a clinical ultrasound system.

B-mode imaging sequences use short pulses (on the order of microseconds) and a PRF in the kilohertz range, while other modes, such as CW Doppler, emit a continuous pulse. Many reports of biological effects or drug delivery have not fully reported all parameters (often reporting only the time-averaged ultrasound intensity) and this uncertainty makes rigorous evaluation of these studies difficult. However, it is clear that successful, fully documented, drug delivery studies employing a long pulse (or high duty cycle) have been reported.

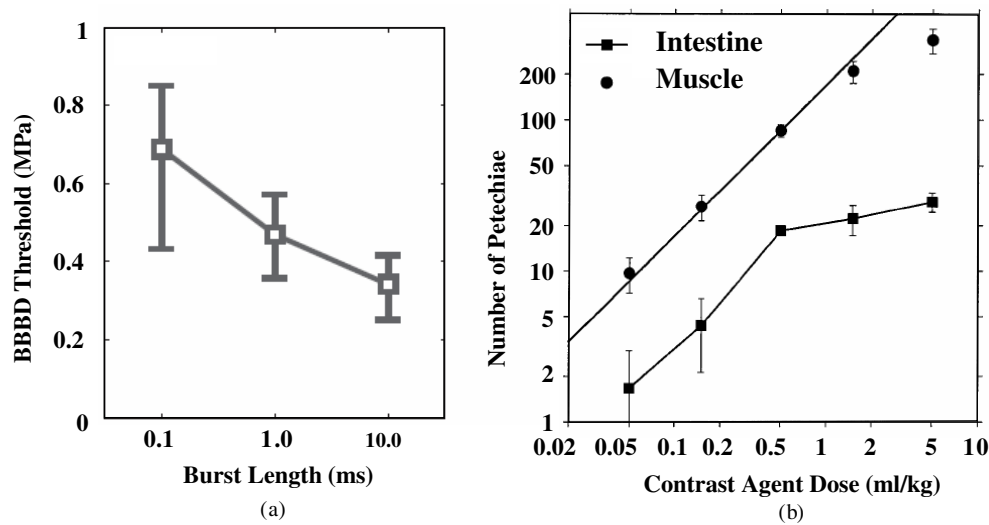


Figure 10. Effect of burst length and contrast agent dose on threshold for biological effects. (a) Effect of burst length on the threshold for blood–brain barrier (BBB) disruption. Reproduced with permission from McDannold *et al* (2008). (b) Occurrence of petechial hemorrhage as a function of Optison dosage in the mouse intestine and muscle. Reproduced with permission from Miller and Qudus (2000).

In studies designed to deliver drugs across the blood–brain barrier (BBB) with a 1.5 MHz center frequency, pulse durations of 10 μ s (with a PRF of 1 kHz) and 10 ms (with a PRF of 10 Hz) were examined (achieving a similar time-averaged acoustic intensity) (Treat *et al* 2007, Hynynen *et al* 2003). The PRP threshold for the 10 μ s pulse was found to be approximately 6.3 MPa, compared to 1.1 MPa for the longer acoustic pulse (Treat *et al* 2007, Hynynen *et al* 2003); therefore, a longer pulse had a greater effect even with an identical duty cycle. These studies were performed in two different species (rat and rabbit); however, the threshold for BBB disruption has previously been reported to be similar between these two species (Hynynen *et al* 2001, Treat *et al* 2007). Furthermore, McDannold *et al* have shown that the threshold for BBB disruption decreases from 0.7 to 0.4 MPa when increasing pulse length from 0.1 to 10 ms (McDannold *et al* 2008) (figure 10(a)).

A similar effect has been noted in our laboratory (Caskey *et al* 2009) where the threshold for disruption of a gel boundary decreased from 2.5 MPa to 1.2 MPa with a pulse duration increasing from 10 μ s to 10 ms. In these experiments, the PRF was modified so that the time-averaged acoustic intensity was matched. Shorter pulses can also be efficacious; many studies have shown successful permeability enhancement using pulses with lengths in the microsecond range (Stieger *et al* 2007, Wible *et al* 2002, Miller and Gies 1998). However, there are advantages to longer pulses as lower acoustic pressures can be used for therapeutic effects, avoiding potential issues associated with high PRPs, such as unintentional cavitation.

5.2.3. Time between destructive pulses. Typically, the estimation of microvascular flow rate uses high-amplitude destructive pulses separated by seconds, together with low-amplitude imaging pulses separated by hundreds of microseconds. In order to accurately estimate flow rate, time for microbubbles to re-fill the region of interest must be allowed between ‘destructive’ pulses.

Wible *et al* showed that a 30-frame-per-second rate decreased renal hemorrhage as compared with a 1-frame-per-second rate (transmitted center frequency of 1.8 MHz, MI of 1.6). At the higher frame rate, microbubbles have insufficient time to reach the capillaries of the kidney and instead are destroyed in larger vessels (Wible *et al* 2002). Similarly, Miller and Quddus found that 10 s of continuous insonation yielded a similar number of petechial hemorrhages as a 100 s continuous insonation; however, hemorrhages increased when they performed 10 insonations with durations of 10 s at a pulse repetition frequency of 0.1 Hz. Thus, although the 10×10 s insonation contained the same number of acoustic cycles as a 100 s insonation, the effect on the surrounding tissue increased. The authors hypothesize that the 10 s delay between pulses allowed for the entry of fresh contrast agents (Miller and Quddus 2000).

5.2.4. Contrast agent size distribution and shell parameters. Increased resting microbubble diameter is hypothesized to increase the interaction of microbubbles with the endothelium (Caskey *et al* 2007). The efficacy of microbubble-enhanced gene delivery has been compared for two albumin-shelled agents, Optison and PESDA. PESDA has a greater fraction of bubbles in the 6–10 μm range (90th percentile at $8.3 \pm 2.9 \mu\text{m}$ for PESDA versus $6.7 \pm 0.7 \mu\text{m}$ for Optison; *P* is not significant), although these agents have similar mean resting radii and shell properties. Using an insonation frequency of 1 MHz, PESDA enhanced gene delivery *in vitro* and *in vivo* to a greater extent than Optison (Pislaru *et al* 2003). Similarly, the threshold for creating petechiae in the mouse intestine is reduced with PESDA compared to Optison; for a center frequency of 2.3 MHz, the threshold for PESDA was 1.8 MPa and the threshold for Optison was 2.2 MPa (Miller and Gies 2000). Hynynen *et al* have compared the efficacy of Definity and Optison for delivering the MRI contrast agent Magnevist beyond the BBB. The pressure threshold for delivery for both agents was 0.4 MPa at a transmission frequency of 0.69 MHz; however, damage to the parenchyma was more prevalent with Optison than Definity using matched acoustic parameters (McDannold *et al* 2007). Comparing Optison and Definity (table 1), Optison has a larger mean diameter; the difference in shell material is not expected to significantly alter the oscillation characteristics. Lipid and albumin-shelled bubbles oscillate with a similar peak expansion and wall velocity, since the shell is very thin and does not constrain expansion for typical ultrasound parameters (Chomas *et al* 2002). In our laboratory, we have observed in *ex vivo* rat cecum that bubbles larger than 4 μm oscillate for 176 ± 139 pulses on average, compared to 31 ± 14 pulses for bubbles smaller than 4 μm (for a transmitted center frequency of 1 MHz, PRP of 800 kPa and 10-cycle pulses). Larger bubbles interacted with the vessel wall for a longer period of time. Bubbles typically required tens to hundreds of 10-cycle pulses to traverse from the center of the lumen of a small vessel to the endothelium wall. Bubbles smaller than 2 μm fragmented prior to reaching the endothelium and were not observed to interact with the vessel wall (Caskey *et al* 2007). We have also observed that microbubbles with diameters of $9.3 \pm 3.4 \mu\text{m}$ are capable of producing tunnels in a gel flow phantom using a transmission frequency of 2.25 MHz, while bubbles outside this size range did not produce visible effects on the gel phantom (Caskey *et al* 2009). If differences due to shell material for albumin and lipid-shelled bubbles are negligible, the studies reviewed here indicate that larger microbubbles decrease the threshold for generating bioeffects.

5.2.5. Contrast agent dosage. An increasing propensity for biological effects with increasing dose is expected since the number of bubbles near boundaries increases with increased concentration. After insonation with a center frequency of 2 MHz, both lethal and repairable

sonoporation of lymphocytes have been reported in a suspension of cells and albumin-shelled microbubbles at various concentrations (Ward *et al* 2000). Sonoporation was observed only for limited separation distances between bubbles and cells and cell lysis reported to increase with decreasing distance between bubbles and cells (Ward *et al* 2000).

In vivo and *ex vivo* studies of contrast-enhanced drug delivery and bioeffects utilize a very wide range of injected contrast dose—spanning $50 \mu\text{L kg}^{-1}$ to 5 mL kg^{-1} (Miller and Quddus 2000, McDannold *et al* 2008). The number of petechial hemorrhages was reported to be approximately proportional to contrast dosage using Optison in mouse intestine and abdominal muscle for a dosage above 1 mL kg^{-1} (figure 10(b)) (Miller and Quddus 2000). In the lower dosage range of $50\text{--}250 \mu\text{L kg}^{-1}$, varying the Optison dose was not reported to significantly increase the MRI signal intensity in experiments designed to deliver gadolinium contrast to the brain of a rabbit (McDannold *et al* 2008). In an *ex vivo* rat cecum, the infusion of a high concentration of microbubbles (>20 microbubbles per $200 \mu\text{m} \times 200 \mu\text{m}$ field) resulted in a small distance between the injected bubbles (Caskey *et al* 2007), as shown in figure 4. In the insonified cecum, the microbubbles then fused and the larger, fused microbubbles interacted with the vessel wall. At lower concentrations, coalescence was not observed and the microbubbles fragmented before extended interaction with the vascular endothelium could occur (Caskey *et al* 2007).

6. Conclusion

The use of microbubble contrast agents to image blood vessel density and perfusion continues to grow, with an increasing number of pulsing strategies having been identified. Combinations of unique transmitted pulses with radiofrequency and pulse-to-pulse filtering result in high target-to-background ratio images. Models for the oscillation and disruption of ultrasound contrast agents have been extensively studied and have been shown to accurately describe the oscillation of microbubble contrast agents for a varied range of parameters. Extensions to accurately model new shell materials and microbubbles located near boundaries have been reported and are expected to be extended. In small animal models, the ratio of pressure to frequency has shown promise to predict the likelihood of capillary hemorrhage; by controlling contrast dose and imaging parameters, such effects should be avoidable. However, the interaction of microbubbles with tissue remains a subject of extensive theoretical and experimental studies at the current time, particularly geared to optimize local drug delivery.

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Corrigendum

Ultrasound contrast microbubbles in imaging and therapy: physical principles and engineering

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In equation (7) the subscript ‘S’ should be included in the denominator of the first term on the right-hand side of the equation to denote the density of the shell. The corrected equation is

$$\begin{aligned} \ddot{R}_1 R_1 \left[1 + \left(\frac{\rho - \rho_s}{\rho_s} \right) \frac{R_1}{R_2} \right] + \dot{R}_1^2 \left[\frac{3}{2} + \frac{\rho - \rho_s}{\rho_s} \frac{4R_2^3 - R_1^3}{2R_2^3} \frac{R_1}{R_2} \right] \\ = \frac{1}{\rho_s} \left\{ p_g(t) - p_0 - p_i(t) - \frac{4\eta R_1^2 \dot{R}_1}{R_2^3} - \frac{2(\sigma_1 + \sigma_2)}{R} \right. \\ \left. - 4 \frac{R_2^3 - R_1^3}{R_2^3 R_1} [\mu_s \dot{R}_1 + G_S(R_1 - R_{10})] \right\}. \end{aligned} \quad (7)$$